

SE LAB3

Architecture Selection & Justification

Household Carbon Footprint & Target Tracker

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Architecture Selection

I have chosen Layered Architecture for the Household Carbon Footprint & Target Tracker system.

Reason 1 — Natural fit for the system's data-processing pipeline

The core use case (Compute Monthly Carbon Footprint, FR-001) follows a strictly sequential pipeline: consumption data is entered and validated, passed to the emissions calculation engine, stored, and then displayed as trends and progress. This maps directly onto Presentation, Business Logic, and Data layers, giving each layer a single, well-defined responsibility and keeping the EPA conversion-factor logic isolated from both the UI and storage concerns.

Reason 2 — Clean support for multiple actor roles

Residents and the Sustainability Coach (FR-005) need different views over the same underlying data (consumption logs, emissions history, targets). A layered design lets both actor journeys share the same Business Logic and Data layers while the Presentation layer exposes role-appropriate screens, avoiding duplicated calculation or storage logic.

Security Advantage

Household consumption and emissions records are sensitive personal data. In a layered architecture, only the Business Logic layer is permitted to access the Data layer directly — the Presentation layer (including the Sustainability Coach's view) must route every request through validated business-logic interfaces (e.g. `ReadEmissionHistory`). This enforces access control and input validation at a single choke point, preventing unauthorised or malformed data from reaching storage, and aligns with NFR-001's requirement to meet security standards while rendering the milestone chart.

Performance Benefit

Because layers are cleanly separated, the Dashboard & Reporting Service can cache or pre-aggregate monthly/annual emissions summaries independently of the calculation engine, so historical trend and milestone charts can be rendered without re-running EPA conversion calculations on every request — directly supporting NFR-001's under-200ms interactive rendering requirement even as consumption history grows.

Component Diagram

